Method Report: EOSC Pillar Qualitative Study

The research question for the qualitative interviews are guided by the following question: What characteristics do researchers value when reusing data? More specifically, we are interested in the aspects that can be incorporated during the technical implementation of EOSC. For instance, we aim at understanding which problems researchers face when accessing data given that experts are currently working on technical aspects of access control.

In order to answer these questions, we conducted semi-structured interviews. We applied the conference tool Microsoft Teams (and Zoom in case Teams did not work) for conducting the interviews. To recreate the atmosphere of a face-to-face interview as closely as possible, we asked participants to use a webcam during the interview. For further analysis we recorded the interviews and transcribed them using a respective software. We asked participants to sign a consent form as precondition for participating (available at the repository).

We drew a non-probability sample using purposive sampling. We determined which characteristics our sample should fulfill to provide the most valuable insights into our research question. We selected respondents based on the following criteria: researchers that reuse data for research, different scientific disciplines, location, career stage, researchers who work at universities or research institutes. We also took gender into account and relation to EOSC-activities.

We only invited researchers who have already considered reusing data for their research. Researchers did not necessarily have to complete this endeavour, but we excluded researchers who categorically rejected the idea of reusing data as we are interested in the specific aspects of reusing data (and not the motivation for reusing data). We sampled researchers according to their scientific disciplines using the OECD criteria: Natural sciences, Engineering and technology, Medical and health sciences, Agricultural sciences, Social sciences and Humanities. In the first research cycle, we interviewed one researcher of each of these six disciplines. In the following research cycles, we incorporated the sub-categories. For instance, if a researcher from the first research cycle representing natural sciences is active in physical sciences, the researcher representing natural sciences in the second research cycle ideally works in a different sub-field, e.g. mathematics. We aimed at including as many sub-disciplines as possible to diversify the views on data reuse, but due to the small sample, were able to cover some sub-disciplines. In addition, we selected researchers based on their work place in one of the five EOSC-Pillar countries (Austria, Belgium, Germany, France, Italy). We interviewed a total of 14 researchers.

Our sampling strategy regarding countries was twofold: On the one hand, we aimed at including researchers from all countries. On the other hand, we acknowledged the different size of the countries by inviting more researchers from the three larger countries. In addition, we took differences within countries into account up to a certain extent. For instance, we acknowledged regional differences by selecting researchers who are affiliated to different organizations in different regions. However, as the sample of a qualitative research by definition is small, we by no means are able to respect all regional differences. As researchers in different career stages may have different perceptions, we took the following career stages into account: PhD students, post-doctoral researchers and tenured professors. The ratio of these categories in our sample should approximately reflect the distribution in reality, hence, more PhD students than post-doctoral researchers and more post-doctoral researchers than tenured professors. We only included researchers who mainly work for universities or research institutes that do not primarily pursue commercial interests. In other words, we excluded researchers who exclusively work for private organizations that pursue commercial interests. We took the gender of researchers into account

when drawing the sample in order to ensure the representation of women. The distribution of women approximately followed the actual distribution of women in the scientific disciplines. As the distribution of women certainly differs across disciplines and career stages, men are most likely over-represented in the sample. In order to get a fresh perspective, we primarily invited researchers who are not involved in open science initiatives or EOSC-related activities.